Exercise 1.1

1. Say whether each of the following is true or false.

(i) 3 is a natural number (ii) $4\frac{1}{2}$ is a natural number

(iii) -2 is a natural number (iv) 101 is a natural number

(v) The smallest natural number is 2.

2. List the numbers in each of the following:

(i) the even natural numbers between 7 and 15

(ii) the odd natural numbers less than 12

(iii) the first four natural numbers

(iv) four consecutive natural numbers beginning with 7

(v) five consecutive odd natural numbers beginning with 5.

3. Describe in words each of the following arrays of numbers:

(i) 1, 2, 3, 4, 5 (ii) 1, 3, 5, 7, 9 (iii) 2, 4, 6

(iv) 6, 8, 10, 12 (v) 9, 11, 13, 15, ... (vi) 101, 102, 103, ...

4. Write down the value of the red digit in each of these numbers:

(i) 38**4** (ii) **4**96 (iii) 6**2**49 (iv) **4**792 (v) 13**4**9

5. Write down the three numbers shown in the diagram below.

Start at the top.

6. Write down

(i) a four-digit number with a 4 in the thousands column.

(ii) a two-digit number with a 3 in the tens column.

(iii) a five-digit number with a 1 in the hundreds column.

(iv) a three-digit number with a 9 in the units column.

(v) a four-digit number with a 0 in the tens column.

(vi) a five-digit number with a 4 in the hundreds column.

(vii) a four-digit number with a 7 in every column except the tens column.

(viii) a five-digit number with a 6 in the thousands column and the units column.

7. Using all the digits 4, 3, 1, 7, what is

(i) the largest (ii) the smallest four-digit number that can be formed from them ?

8. What is the difference between the values of the two fours in the number 34746?

9. What is the sum of the values of the threes in the number 34353?

10. What is the difference between the largest and smallest four-digit numbers that can be formed using the digits 2, 9, 3, 5?

11. Write down each of these numbers:

(i) seven hundred and nine
(ii) two thousand, three hundred and nine
(iii) six thousand and four
(iv) twenty thousand, four hundred and two
(v) ninety thousand and ninety
(vi) three hundred and two thousand and forty five
(vii) one million and thirty thousand

12. Write these numbers in order, starting with the smallest:

3297 2988 3079 3190 931 3098

13. Write the next car registration number for each of these:

(i) 10 D 799 (ii) 11 KE 1019 (iii) 12 KY 7999

14. By placing all these cards in a line, 3 8 2 2 1 6, write down

(i) the largest number that can be made

(ii) the smallest number that can be made.

15. Place each of the following lists of number in ascending order:

(i) 604, 406, 46, 0, 64.
(ii) 210, 85, 219, 26, 401, 79, 151.

16. Write down a list, in descending order, of the 5 largest numbers that can be formed using the digits 4, 1, 9, 2.

17. Place each of the following numbers in ascending order;
1121, 2211, 2112, 1212, 1211, 1122.